

Assignment #2

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Objectives

In this assignment, we review the knowledge we learned from Chapter 4 - Probability lectures.

1) Probability sample space (21)

Determine sample spaces S for the following random experiments:

- 1.1) Toss a coin once and record whether you get heads (H) or tails (T).
- 1.2) Toss a coin twice.
- 1.3) Toss a coin 3 times.
- 1.4) Select a person at random and check his or her blood type.
- 1.5) Roll a fair dice (a dice having 6 faces of 1,2,3,4,5,6 dots)
- 1.6) A pair of fair dice is rolled, and the sum of the dots on the two faces that come up is recorded.
- 1.7) A couple decides to have children until they have one boy and one girl, but they will not have more than three children.

2) Disjoint events; independent events (16)

Study the following random experiment (situation) and answer the questions.

Situation-1:

A music class consists of 4 male students and 4 female students. Two of the 8 students are chosen at random for a role-playing exercise.

Let denote events:

A: the first chosen is male.

B: the second chosen is female

- 2.1) are the two above events A and B are disjoint or not disjoint? Explain why?
- 2.2) are the two above events A and B independent or dependent? Explain why?

Situation-2:

Roughly 5% of Canadian males have some sort of color blindness. Suppose that two Canadian males are selected at random for a medical study.

Let C represent the event: “the first male is color blind”

Let D represent the event: “the second male is not color blind”

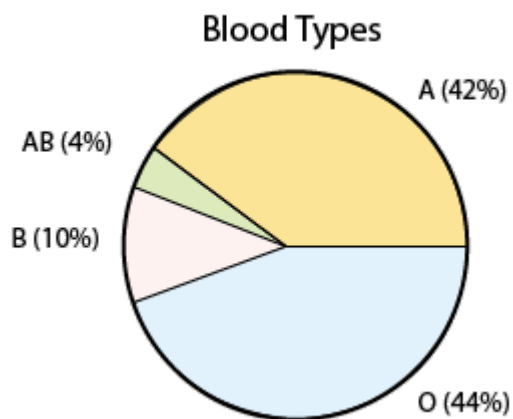
2.3) are the two above events C and D disjoint or not disjoint? Explain why?

2.4) are the two above events C and D independent or dependent? Explain why?

3) Probability rules (18)

Based on the empirical methods and probability rules, answer the following probability questions.

Situation: revisit the blood type example from the Chapter-4 lecture.



Blood type	O	A	B	AB
Probability	0.44	0.42	0.1	0.04

Note: A person with type A can donate blood to a person with type A or AB, type B can donate blood to a person with type B or AB, type AB can donate blood to a person with type AB only, type O blood can donate to anyone.

3.1) What is the probability if a randomly chosen person in the USA does not have blood type AB?

3.2) What is the probability that a randomly chosen person is a potential donor for a blood type B person?

3.3) What is the probability that a randomly chosen person is a potential donor for a blood type AB person?

3.4) Two people are selected at random from all people in the United States. What is the probability that both have blood type B?

3.5) Two people are selected simultaneously and at random from all people in the United States. What is the probability that both have blood type O?

3.6) Two people are chosen simultaneously and at random. What is the probability that both have the same blood type? ($P(\text{same blood type}) = P([O1 \text{ and } O2] \text{ or } [A1 \text{ and } A2] \text{ or } [B1 \text{ and } B2] \text{ or } [AB1 \text{ and } AB2])$)

4) Conditional probability and independence check (12)

Situation: A homeowner has smoke alarms installed in the dining room (adjacent to the kitchen) and an upstairs bedroom (above the kitchen). The two-way table below shows **probabilities** of smoke in the kitchen triggering the alarm in the dining room (D) or not, and in the bedroom (B) or not.

	B	not B	Total
D	0.38	0.57	0.95
not D	0.02	0.03	0.05
Total	0.4	0.6	1.0

4.1) Using the above two-way probability table, find $P(D | B)$?

4.2) Using the above two-way probability table, find $P(D | \text{not } B)$?

4.3) Use conditional probability, check whether the events D and B are independent or not?

5) Normal random variables (15)

Situation: The hourly salary rate for consultants at the company “ABC Consulting Firm” follows a normal distribution, with a mean of \$28 and a standard deviation of \$6.

5.1) Kyle earns \$23.5 an hour. How far is Kyle’s hourly rate far from the mean or what is his z score?

5.2) Sherry also works for “ABC Consulting Firm” and the z-score for her hourly salary is 2.6. What is Sherry’s actual hourly salary?

5.3) Tom also works for “ABC Consulting Firm” and the z-score for his hourly salary is -1.3. What is Tom’s actual hourly salary?

6) Sampling distribution: parameters vs. statistics (16)

Situation: The length of human pregnancies has a mean of 266 days and a standard deviation of 16 days. A random sample of 20 pregnant women was observed to have a mean pregnancy length of 270 days, with a standard deviation of 12 days.

6.1) What are the population parameters from this situation?

6.2) What are the sample statistics from this situation?

Situation: Approximately 60% of all part-time university students in Canada are female. In a study at the ABC university, a sample of 300 part-time students were randomly selected, and within 300 students, there were 155 female students.

6.3) What is the population parameter from this situation?

6.4) What is the sample statistic from this situation?

Submission

- Document your answers for all questions and save as a pdf file.
- Upload your final pdf file to Nexus for grading